

## Meaning and Rationale of $C_{Roads\_D}$

The road density variable, **PROP\_PRIM\_SEC\_ROADS** (Proportion of Primary and Secondary Roads), serves as a fundamental measure of the built environment's capacity to support mobility, often acting as a proxy for walkability or vehicular efficiency.

| Aspect                         | Rationale                                                                                                                                                                                                                                                          |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mobility Proxy</b>          | In rural and micropolitan areas like Decatur County, this variable indicates the <i>quality and density of the network</i> needed for driving, cycling, or walking—especially where transit is absent. Low density means poor infrastructure and reduced mobility. |
| <b>Walkability Proxy</b>       | High density of primary/secondary roads generally suggests a more interconnected, grid-like network, which correlates with better walkability than low-density, sprawling, or disconnected road systems.                                                           |
| <b>Vulnerability Direction</b> | For this component, low road density is treated as high vulnerability. If there are fewer primary/secondary roads, access is inherently more difficult.                                                                                                            |

## Standardization and Direction

The  $C_{Roads\_D}$  component is calculated using the global Z-score standardization method, ensuring that a higher score means higher vulnerability:

1. **Z-Score:** The raw **PROP\_PRIM\_SEC\_ROADS** value is standardized using the global mean ( $\mu$ ) and standard deviation ( $\sigma$ ).
2. **Directional Correction:** Since high road density is a protective factor (low vulnerability), the resulting Z-score is reversed (multiplied by -1).

$$C_{\text{Roads\_D}} = -Z_{\text{PROP\_PRIM\_SEC\_ROADS}}$$

### Unweighted Final Score (For Comparison)

In the unweighted model,  $C_{\text{Roads\_D}}$  is simply added with a weight of 1.0, treating the infrastructure deficit as equally important as the other components:

$$V_{\text{final\_UW}} = C_{\text{Eco}} + C_{\text{Geo}} + C_A + C_B + C_{\text{Internet\_C}} + C_{\text{Roads\_D}}$$

$$C_{\text{Roads\_D}} = -Z_{\text{PROP\_PRIM\_SEC\_ROADS}}$$

The transport score (TVS) is calculated as:

$$\text{TVS} = w_A C_A + w_B C_B + w_C C_{\text{Internet\_C}} + w_D C_{\text{Roads\_D}}$$

$$V_{\text{final\_W}} = C_{\text{Eco}} + C_{\text{Geo}} + \text{TVS}$$

**RUCA (Rural-Urban Commuting Area) codes** classify census tracts based on how people live and travel. Unlike county-level categories, RUCA identifies whether a tract is urban, suburban, rural, or remote by looking at commuting patterns and local connectivity. It gives a more realistic picture of access conditions, especially in areas where a single county includes both dense urban centers and isolated rural communities.

RUCA helps interpret food access barriers in context rather than treating all “rural” or all “urban” places the same.

### Component Methodology Breakdown

All standardization uses the **global mean**  $\mu$ ) and **standard deviation** ( $\sigma$ ) derived from the master dataset, ensuring the vulnerability score for Decatur County is measured against the broader contextual landscape.

Scores derived from global Z-score standardization, Decatur County tracts (N=8)

Model Formula:

- $V_{\text{final\_Weighted}} = C_{\text{Economic}} + C_{\text{Geographic}} + \text{TVS}$
- $\text{TVS} = 0.4 * C_{\text{Vehicle}} + 0.3 * C_{\text{Transit}} + 0.2 * C_{\text{Internet}} + 0.1 * C_{\text{Roads}}$
- $C_{\text{Transit\_B}} = +3.0$  (Systemic Transit Penalty - no public transit in county)

$$\text{TVS} = w_A C_A + w_B C_B + w_C C_{\text{Internet\_C}} + w_D C_{\text{Roads\_D}}$$

### Unweighted Final Score (For Comparison)

In the unweighted model,  $C_{Roads\_D}$  is simply added with a weight of 1.0, treating the infrastructure deficit as equally important as the other components

$$V_{final\_UW} = C_{Eco} + C_{Geo} + C_A + C_B + C_{Internet\_C} + C_{Roads\_D}$$

### 1.) Economic Vulnerability ( $C_{Economic}$ )

| Aspect         | Calculation (Formula)                                                          | Rationale                                                                                                                                                                                                        |
|----------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Variables Used | <b>Income</b> (Median Household Income) and <b>poverty_pct</b> (Poverty Rate). | A single metric (like median income) is insufficient and can mask high poverty. This composite score prevents <b>economic flattening</b> by factoring in both affluence (income) and deprivation (poverty rate). |
| Scoring        | $C_{Economic} = (-Z_{Income}) +$                                               | <b>Low Income</b> is a vulnerability (reversed Z-score). <b>High Poverty</b> is a vulnerability (standard Z-score). The scores are summed to give a single, robust economic measure.                             |

### 2.) Geographic Vulnerability ( $C_{Geo}$ )

| Aspect         | Calculation (Formula)                                                                                                                                                                                         | Rationale                                                                                                                                                                                                     |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Variable Used. | <b>den_totalfoodstores</b> (Density of total food stores).Quantifies physical access to food resources beyond the binary "food desert" definition. This is a direct measure of <b>resource availability</b> . | <b>den_totalfoodstores</b> (Density of total food stores).Quantifies physical access to food resources beyond the binary "food desert" definition. This is a direct measure of <b>resource availability</b> . |
| Scoring        | $C_{Geo} = -Z_{den\_totalfoodstores}$                                                                                                                                                                         | <b>High Density</b> is a protective                                                                                                                                                                           |

|  |  |                                                                                                                                   |
|--|--|-----------------------------------------------------------------------------------------------------------------------------------|
|  |  | factor (low vulnerability), so the Z-score is <b>reversed</b> . A low-density tract receives a high positive vulnerability score. |
|--|--|-----------------------------------------------------------------------------------------------------------------------------------|

### 3.) Transport Vulnerability: Vehicle Access ( $C_{Vehicle\_A}$ )

| Aspect         | Calculation (Formula)                                                   | Rationale                                                                                                                                           |
|----------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Variable Used. | Vehicles_PctHH_NoVehicle<br>(Percentage of Households with No Vehicle). | Measures dependence on non-existent or unreliable transport infrastructure. In rural settings, this is often the single greatest barrier to access. |
| Scoring        | $C_{Vehicle\_A} = Z_{Vehicles\_PctHH}$                                  | <b>High Percentage of No-Vehicle Households</b> is a vulnerability, so the Z-score direction is <b>kept positive</b> .                              |

### 4.) Transport Vulnerability: Transit Access ( $C_{Transit\_B}$ )

| Aspect         | Calculation (Formula)     | Rationale                         |
|----------------|---------------------------|-----------------------------------|
| Variable Used. | STOPS_PER_CAPITA (Transit | Measures public transport service |

|                |                                                                                     |                                                                                                                                                                                                                                                                                                                      |
|----------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | Stops per Capita                                                                    | availability.                                                                                                                                                                                                                                                                                                        |
| <b>Scoring</b> | $C_{\text{Transit}_B} = -Z_{\text{STOPS\_PER\_CAPITA}}$<br>with <b>Imputation</b> . | <b>High Stops</b> is a protective factor (reversed Z-score). For tracts with <b>zero transit availability (NaNs)</b> , the standardized Z-score is imputed to $\mathbf{+3.0}$ to reflect a <b>Systemic Absence Penalty</b> (Extreme Vulnerability), preventing the erasure of this critical infrastructural deficit. |

#### 5.) Transport Vulnerability: Internet Access ( $C_{\text{Internet}_C}$ )

| Aspect                | Calculation (Formula)                                                                                                  | Rationale                                                                                                                                                                                                               |
|-----------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Variable Used.</b> | <code>est_NO_INT</code> , <code>pct_no_internet</code> , and <code>pct_cellular_broadband</code> .                     | Captures multiple facets of digital exclusion. Accessing transit can be entirely data access dependent. Digital connectivity acknowledges this contextual reality. Lack-of-access increases lack of mobility potential. |
| <b>Scoring</b>        | $C_{\text{Internet}_C} = Z_{\text{est\_NO\_INT}} + Z_{\text{pct\_no\_internet}} - Z_{\text{pct\_cellular\_broadband}}$ | The component is the sum of two <b>Lack of Access</b> variables (kept positive) minus one <b>Access</b> variable (reversed/subtracted), creating a single robust digital vulnerability score.                           |

#### 6.) Transport Vulnerability: Roads/Walkability ( $C_{\text{Roads}_D}$ )

| Aspect         | Calculation (Formula)                                               | Rationale                                                                                                                                                                                 |
|----------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Variable Used. | PROP_PRIM_SEC_ROADS<br>(Proportion of Primary and Secondary Roads). | Serves as a <b>proxy for mobility infrastructure and walkability</b> .                                                                                                                    |
| Scoring        | $C_{\text{Roads}_D} = -Z_{\text{PROP\_PRIM\_}}$                     | <b>High Road Density</b> is a protective factor, so the Z-score is <b>reversed</b> . This component allows for identification of the <b>infrastructure deficit</b> component of mobility. |

## Global Scoring Methodology

The final scoring combines these components into two primary indices, allowing for the critical comparison that demonstrates the need for contextualization.

### A. Unweighted Final Score ( $V_{final\_UW}$ )

This score treats all components as equally important (weight of 1.0) and serves as the baseline for comparison.

$$V_{final\_UW} = C_{Eco} + C_{Geo} + C_A + C_{Transit\_B} + C_{Internet\_C} + C_{Roads\_D}$$

### B. RUCA-Weighted Final Score ( $V_{final\_W}$ ) (The Authoritative Score)

This score incorporates the **Context-Sensitive Weighting** required for equity-informed analysis. The Decatur County tracts (designated Micropolitan) are mapped to the "**Large Town**" weighting matrix, which down-weights transit and up-weights vehicle access.

$$V_{final\_W} = C_{Eco} + C_{Geo} + \text{TVS}$$

The final comparison of  $V_{final\_UW}$  and  $V_{final\_W}$  forms the core argument of this project: the weighted model is necessary to prevent the **erasure** of localized, high-impact transport barriers.

All of these scoring matrices are imperfect and meant for refinement, critique and serve as a starting point. The [USDA definition](#) of “food desert” was the impetus for this project and my takeaway from the definition (though refinements regarding its limitations have been improved that has not manifested in real world application), in plain language is that a food desert is static, binary and measured by two variables; income and proximity to a food resource.

### ### Standardization via Z-Scores

This model standardizes component variables using Z-scores to place heterogeneous measures on a common scale prior to aggregation. This choice is pragmatic and problematic. I do not pretend that it is not "inferential" because it carries several explicit assumptions and limitations. I don't like it either and would very much like an alternative.

Z-score standardization assumes that variables are meaningfully centered around a global mean and scaled by a global standard deviation. In classical applications, this approach is often paired with assumptions of approximate normality, independence of observations, and large-sample behavior (via the Central Limit Theorem). These conditions are **\*\*not fully met\*\*** in this context.

First, the component variables are not assumed to be independent. Transportation access, economic conditions, infrastructure, and connectivity are structurally entangled by design. Z-scores are used here to standardize magnitude, not to imply orthogonality or causal separability among components.

Second, normality is not assumed. Several variables exhibit skewed or bounded distributions, particularly in rural or micropolitan contexts. Z-scores are employed as a relative scaling mechanism rather than as a probabilistic model of the data-generating process.

Third, this analysis does not rely on asymptotic inference. The use of Z-scores is not intended to support hypothesis testing, confidence intervals, or population-level

generalization. Instead, standardization enables within-model comparison of relative position and contribution across components.

To mitigate small-sample distortion, all Z-scores are calculated using global reference distributions rather than tract-level or county-level means. This anchors local observations to a broader structural context and avoids artificially inflating variance in small geographic samples.

In this model, Z-scores function as a **\*\*comparability device\*\***, not a claim about statistical independence, normality, or causal structure. Their use is justified insofar as the model's purpose is exploratory, comparative, and diagnostic rather than predictive or inferential.

### ### Assumption: Completeness and Accuracy of Grocery Store Data

This model assumes that grocery store locations drawn from publicly available datasets represent a sufficiently complete approximation of the food retail landscape for the purposes of structural comparison. Store presence is treated as binary and exhaustive: a store either exists in the dataset or it does not.

This assumption is knowingly fragile. While multiple data sources were cross-checked, the model does not independently verify store operational status, food quality, affordability, cultural relevance, or temporal stability. Stores may be missing, misclassified, closed, inaccessible, or functionally unusable for reasons not captured in spatial data.

Crucially, this uncertainty is **\*\*not corrected for within the scoring framework\*\***. No weighting, penalty, or confidence adjustment is applied to account for incomplete or inaccurate store representation. As a result, geographic density and availability signals should be interpreted as provisional structural indicators rather than confirmed access conditions.

This assumption reflects a broader constraint of food access research: the absence of reliable, continuously updated ground-truth data on food availability. The model treats grocery access as “present and accounted for” only in the narrow sense that locations are mapped, not in the substantive sense that food is accessible, affordable, or usable.